MASFE: An Onboard Multi-Algorithm Scheduling and Fusion Engine

For Regenerative Agriculture SmallSat Missions

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Solo submission

Abstract. MASFE moves regenerative-agriculture monitoring onboard by screening every pass with lightweight EVI/NDWI cues, escalating likely stress through CSC fusion, and downlinking only alert tiles. In a 100-seed Monte Carlo benchmark, MASFE achieves 98.9% downlink reduction versus raw, 38.3% energy savings, 100% disease-event recall, and a 1.6% false-positive rate within the selected payload budget. A 2024 Westlands/Firebaugh MODIS replay exercises the policy on real scenes (§3). The critical path to TRL 4 is hardware-in-the-loop validation (§5).

1 Introduction

Regenerative agriculture needs crop-stress detection early enough for patch-level intervention instead of blanket chemical application [1, 2, 3, 4]. Current land products are often too slow or coarse for that workflow.

This work addresses that gap by introducing onboard, belief-driven sensing via a finite-horizon MDP-based scheduler; Fig. 1 and §2 give the full pipeline and implementation.

Table 1: MASFE response to all five judging criteria.

Criterion	**Where**	**Key evidence**
Creativity	§2–3	EO-1 autonomy heritage; contrast to ASPEN/CASPER; complementary to Dynamic Targeting (§2).
Feasibility	§2–3	SWAP closure, compute margin, and real-scene replay.
Impact	§3	Monte Carlo reduction, recall, and false-positive evidence.
Business	§4	5-year rollout and multi-platform distribution.
Presentation	Eq. (1), Fig. 1	repository link; architecture diagram; CSC equation; reproducible simulation submission.

2 Proposed Solution Detail

MASFE uses a two-stage adaptive sensing loop. Stage 1 executes an EVI anomaly screen derived from MOD13A1 logic on every observed field patch and updates a per-patch stress posterior using EVI and NDWI cues from the MSI pass. When the posterior indicates likely stress, Stage 2 runs a thermal retrieval derived from MOD11A1 and fuses EVI, LST, and MOD09A1-derived NDWI into a Crop Stress Composite (CSC), a normalized anomaly index conceptually related to temperature–vegetation dryness indices (e.g., TVDI) [3, 4, 5, 6]. Only confirmed anomalous tiles are buffered for priority downlink. This makes sensing content-driven rather than timeline-driven and distinguishes MASFE from ASPEN/CASPER-style planners, which optimize scheduled activities rather than per-patch stress belief [7, 8]. EO-1 ASE demonstrated rule-based change-detection triggers in orbit [9]; MASFE advances that heritage with per-patch Beta posteriors that accumulate evidence across passes, enabling stress-trajectory-sensitive scheduling rather than single-pass exceedance detection. Candela et al.'s Dynamic Targeting instead focuses on rapid retargeting of pointable instruments toward transient high-value scenes; MASFE is complementary in operating at composite cadence for chronic crop stress monitoring [10].

$$\text{CSC} = 0.40 \cdot \text{clip}\left(\frac{\Delta\text{EVI}}{5\sigma_e}, 0, 1\right) + 0.35 \cdot \text{clip}\left(\frac{\Delta\text{LST}}{4\sigma_l}, 0, 1\right) + 0.25 \cdot \text{clip}\left(\frac{\Delta\text{NDWI}}{4\sigma_n}, 0, 1\right) \quad (1)$$

where $\Delta\text{EVI} = \max(0, \text{EVI}_{base} - \text{EVI}_{obs})$, $\Delta\text{LST} = \max(0, \text{LST}_{obs} - \text{LST}_{base})$, and $\Delta\text{NDWI} = \max(0, \text{NDWI}_{base} - \text{NDWI}_{obs})$. The factors of 5, 4, and 4 are heuristic saturation divisors chosen so CSC terms map typical benchmark anomalies into $[0, 1]$ before composition; crop-specific

agronomic recalibration is a pre-TRL-4 item.
The scheduling logic is modeled as a finite-horizon MDP with per-patch Beta posteriors Beta(α_i, β_i) over stress probability that accumulate evidence across passes. Stage 1 increments α_i when EVI or NDWI exceedance is observed and β_i otherwise, with the evidence window capped to preserve responsiveness. The deployed rule-based controller maps patchwise beliefs into a compact summary $(\mu_{\max}, \tau_{\max}, \mathrm{SOC}, d, k)$ for exposition, where $\mu_{\max}$ is the maximum posterior mean, $\tau_{\max}$ is the maximum $P(p > 0.5)$ tail probability, SOC is battery state of charge, $d \in \{0, 1\}$ indicates downlink availability, and k counts passes since last fusion. Among these, **action selection** in code uses $\tau_{\max}$, SOC, and k (battery tiers and exploration cadence); $\mu_{\max}$ and d enter the separate `FUSE`→`FUSE_PRIORITY` promotion gate for alert-tile export under a duty-cycled contact window. The action set remains {SKIP, MOD13, FUSE, FUSE_PRIORITY}; NDWI is derived from the same MSI acquisition rather than requiring a separate action. In the benchmark, the tiered resolution pyramid is modeled as action metadata: MOD13 corresponds to 30 m screening, FUSE corresponds to a 10 m confirmation tier, and FUSE_PRIORITY corresponds to 4.6 m native alert evidence; the 10 m tier is the confirmation-resolution metadata attached to FUSE, not a separate fifth action. The deployed solver is a hand-tuned deterministic policy chosen for flight-V&V traceability.

Table 2: Mission requirements and demonstrated MASFE results.

Requirement	**MASFE response**
Alert latency (observation pass to first ground contact) $\leq$ 90 min	One-pass onboard screening and fusion followed by X-band priority downlink at the first available contact.
Spatial resolution $\leq$ 30 m GSD	Tiered GSD (30/10/4.6 m) tied to MOD13, FUSE, and FUSE_PRIORITY action metadata (§2).
False-positive rate $\leq$ 15%	1.6% with benign confounders and 36.7 cloud-obscured passes per seed on average ($\sim$31% of 120 passes; §3).
SmallSat SWAP compliance	Power: 9.6 W peak / 5.05 W duty-cycled payload average Compute: 92.5% peak LEON4FT ceiling; 39.2% Stage-1; 49.2% seasonal average Energy: 86% observed benchmark minimum SOC in local 40 Wh buffer.
Downlink efficiency	Large reduction versus raw downlink with full disease-event recall; point estimates and CIs in §3.

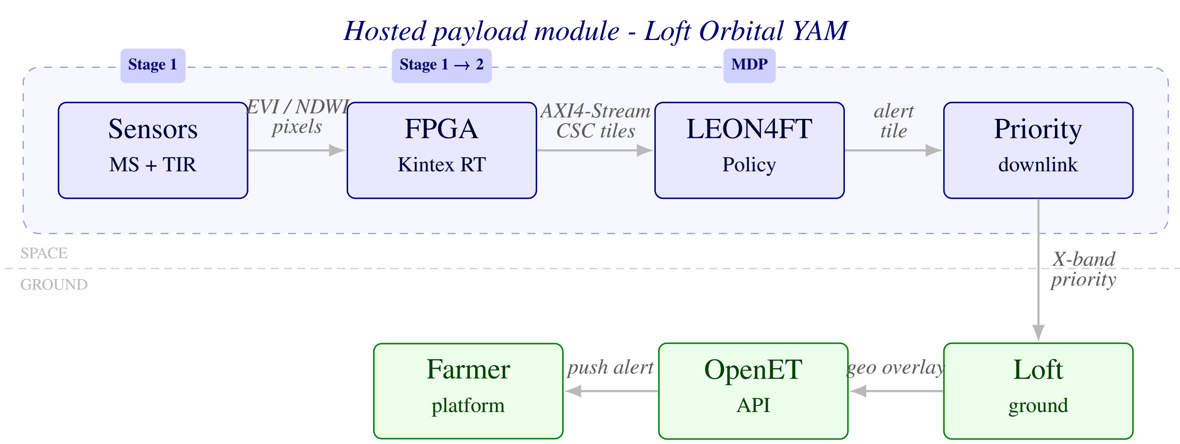

Figure 1: MASFE end-to-end architecture: hosted payload (dashed) on Loft Orbital YAM from sensors through priority X-band downlink; ground path to farm software via OpenET (full sensing loop, §2).

The FPGA performs radiometric preprocessing and per-pixel CSC computation, then transfers frames to the LEON4FT over AXI4-Stream (budgeted under 1 ms). CSC is evaluated as a pass-to-pass relative anomaly detector rather than a fully atmosphere-corrected retrieval. The selected Kintex UltraScale+ RT part comes from AMD's radiation-tolerant XQR family [11]. MASFE is framed as a hosted payload module within a Loft Orbital YAM bus; the SWAP budget therefore describes the payload module itself, not the host spacecraft.

Operational degradation is explicit. Between 15% and 35% battery state of charge, the degraded mode preserves Stage 1-only screening; below 15%, it preserves alert storage and downlink. This satisfies the challenge requirement to degrade gracefully under onboard resource stress [1].

Processor utilization remains below the worst-case ceiling and preserves margin for RTOS and fault-handling overhead.

Each alert packet contains a geo-tagged field-patch polygon with centroid coordinates, CSC severity score, contributing EVI/LST/NDWI anomaly components, timestamp and pass ID, and a confidence flag derived from consecutive confirming passes. Escalated alert tiles are compressed onboard using CCSDS 122.0 wavelet coding with a planning assumption of roughly 2.5:1, reducing a nominal 8.4 MB/km^2 native alert stream to about 3.36 MB/km^2 delivered before packetization [12]. Per-patch geolocation is derived from bus ephemeris and attitude solutions and is budgeted to remain comfortably inside the 30 m effective-GSD requirement. These packets are delivered through Loft Orbital's ground network to farm software as geo-tagged overlays via the OpenET-linked API, enabling push notifications within 15 minutes after first ground contact.

Full simulation source code is available as Code Component 3 at github.com/emillambert/Emil_MASFE.

3 Potential Impact

Dr. Katie Gold's disease-detection work emphasizes that spectral stress often appears before visible symptoms and before whole-field treatment would be agronomically or economically rational [2]. Performance is summarized in Table 3; key distributions and confidence intervals are detailed below. In a 100-seed Monte Carlo benchmark, MASFE was tested over 120 passes, 500 field patches, 21 disease events, and 9 benign confounders per seed, with 36.7 cloud-obscured passes per seed on average. Across that benchmark, the belief gate routes $\sim$81% of passes to Stage-1 screening only, $\sim$17% to FUSE confirmation, and $\sim$2% to FUSE_PRIORITY alert export.

95% confidence intervals for those runs were 98.92–98.97% for downlink reduction versus raw, 38.17–38.34% for energy saving versus raw, 100.0–100.0% for disease-event recall, and 1.34–1.78% for false-positive rate. The downlink-reduction interval is tight because the raw baseline is fixed and MASFE's total byte count is dominated by stable Stage-1 screening traffic, so seed-to-seed variability mainly changes alert counts rather than total data volume. The operating point is tunable: sweeping csc_alert_thr from 0.40 to 0.65 preserved 100.0% recall while moving false-positive rate from 2.23% to 1.26%; at the sweep endpoint 0.70 recall falls to $\sim$98.5% as the policy hits battery-driven conserve behavior, and a $\pm30\%$ CSC-parameter sweep kept recall at 100.0% with false-positive rate in the 1.0–2.3% band; detailed curves and cases are included in the code submission.

Table 3: Baseline comparisons under the synthetic Monte Carlo benchmark.

Variant	Downlink↓	Recall↑	FP↓	Compute↓
Raw downlink	0.0%	100.0%	1.6%	N/A
Fixed-period onboard	94.8%	100.0%	1.6%	92.5%
EVI-only threshold	96.2%	100.0%	2.3%	39.2%
No-belief MASFE	95.1%	100.0%	1.7%	76.2%
Full MASFE	98.9%	100.0%	1.6%	49.2%

Raw downlink denotes no onboard triage; recall and false-positive rate apply the same detector after ground-processing the full dataset, so the 1.6% FP in full MASFE matches the ground-processing detector floor rather than an onboard precision gain. The $\sim$0.1 pp FP gap versus no-belief MASFE reflects the NDWI-inclusive fusion path and belief-gated scheduling under the same post hoc detector. The downlink-reduction gap versus no-belief is driven in part by no-belief issuing more `FUSE_PRIORITY` passes (larger per-pass byte budget for alert tiles) under the benchmark contact model.

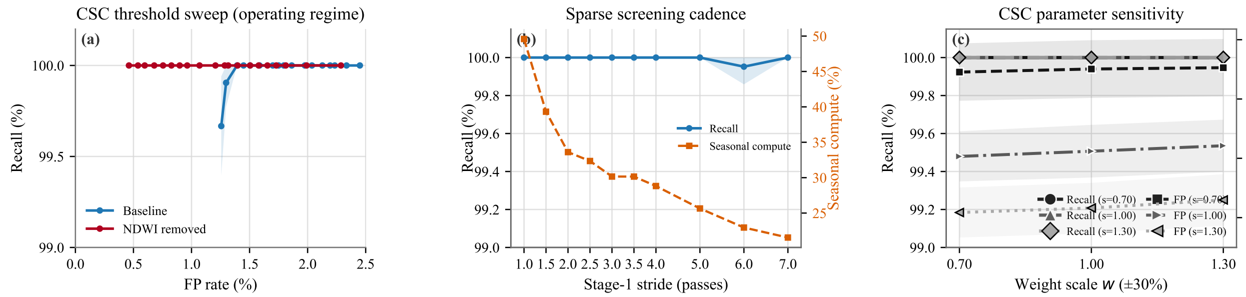

Figure 2: Additional ablations and curves (95% CI bands where applicable). Left: operating-point sweep comparing baseline CSC to an NDWI-removed fallback over the relevant FP band; more permissive thresholds eventually drive recall collapse through battery-driven CONSERVE mode. Middle: sparse Stage 1 cadence, where recall stays near ceiling while *seasonal-average compute* (right axis) shows the cost of under-sampling. Right: ±30% CSC weight/saturation grid (w=weight scale, s=saturation scale); recall stays within 99–100% while *FP rate* (right axis) varies across calibration corners.

The code component also replays official MOD13A1 and MOD11A1 subsets over Westlands/Firebaugh, California, from June through October 2024, aligning MOD13A1 composite dates to a 1 km grid, collapsing MOD11A1 to median clear-sky daytime LST over the same 16-day window, and applying the unchanged MASFE policy without a labeled disease benchmark. The replay path accepts MOD09A1-derived NDWI where available, but the tracked Westlands anchor bundle predates that extension and therefore falls back cleanly to EVI/LST fusion on the cached artifacts. Across 7 valid composite windows after a 3-window warmup, MASFE issued one confirmed FUSE_PRIORITY alert on July 27, 2024 and six additional FUSE-only windows at 99.5% mean valid coverage.

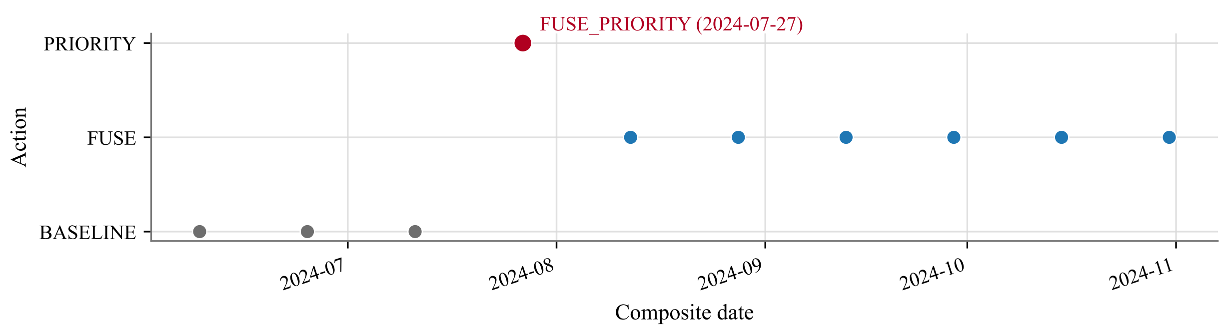

Figure 3: Action timeline for the Westlands replay: MOD13 baseline, FUSE windows, and the single FUSE_PRIORITY alert on July 27, 2024 (setup in §3).

The farm-scale value proposition is concrete. For a 500 ha regenerative tomato operation, UC Cooperative Extension processing-tomato budgets place representative disease-control line items in the rough $74–119/ha/season range, implying roughly $34,000–$55,000/season of avoidable chemical spend if MASFE confines treatment to the roughly 8% of the field flagged during an event [13]. That matters because late blight remains a destructive tomato disease that can destroy entire fields and collapse plants within 7–10 days under cool, wet conditions [14]. Against a global fungicide-and-bactericide baseline of roughly 0.8 Mt active ingredient per year in 2023, MASFE's patch-level triage model changes how fungicide demand is generated, not just farm-level cost [15].

4 Team Experience and Market Evaluation

Emil Wes Lambert is completing a BSc in Aerospace Engineering at TU Delft, with coursework in spacecraft systems, AOCS, mission analysis, and embedded flight software. At AeroDelft he has led partnerships and business operations, including procurement, sponsor relations, and participation in the EUR 73 M HAPSS hydrogen-propulsion consortium alongside Airbus, KLM, and NLR. That background supports MASFE's SWAP realism, AIST-track funding strategy, and commercial positioning.

California is the validation anchor and initial go-to-market wedge. The same architecture scales from SJV into the wider U.S. irrigated base, then into EU and Brazilian irrigation markets, and finally into a global irrigated-farmland platform with the same onboard scheduler and region-specific recalibration of CSC thresholds [16, 17, 18, 19]. The paper-facing economics therefore use a low/design-to-cost case, a \$4/ha/year alerting product, a modeled 90% annual retention rate, and a Year 2 soil-carbon MRV add-on aligned to Verra VM0042-style workflows [20].

Table 4: Illustrative MASFE rollout from SJV pilot to global platform.

Year	Milestone	Geography	Coverage	Revenue	Op. margin
Y1	SJV pilot	California SJV	3.5% SJV	\$0.20 M	grant-funded
Y2	California scale	California	28.3% SJV	\$1.70 M	17.1%
Y3	U.S. national	U.S. irrigated	11.0% US	\$10.62 M	60.4%
Y4	EU + Brazil entry	EU + Brazil	35.1% EU+Brazil	\$27.62 M	68.1%
Y5	Global platform	Global	6.0% global	\$76.50 M	70.7%

In this rollout, Y1 is a paid pilot; Y2 scales through Climate FieldView API integration plus direct enterprise sales; Y3 targets U.S.-scale profitability via Climate FieldView and John Deere Operations Center; Y4 expands through xarvio and John Deere Brazil; and Y5 targets global multi-platform distribution that could also include CropX. These platforms are API integration targets rather than committed channels, and no agreements are yet in place. Y1 is intentionally modeled as an AIST- or NASA SBIR/STTR-style co-funded bridge-to-MVP year rather than a self-financing steady state [21, 22]. In the retained fixed-cost sensitivity, the low case breaks even at Y2 and 400 kha, while the base and high cases still break even at Y3 and 2.5 Mha.

Agriculture is the entry market, but the scheduler can also be retuned for wildfire fuel-moisture monitoring in a broader fire-detection market that one industry estimate projects at about \$55 B by 2030 [23]; flood-front tracking would swap CSC for water-extent products, and urban heat-island response would replace farm polygons with city blocks while reusing the anomaly-driven scheduler.

5 Expected Next Steps

MASFE is currently a TRL 3 concept: the scheduling logic, fusion equation, SWAP closure, and real-scene replay are demonstrated in software, but the payload has not yet been exercised on target hardware. The next milestone is a TRL 4 hardware-in-the-loop demonstration on a LEON4FT-class platform under NASA AIST using the replayed EarthData scene set. That work should also add FPGA-side Fmask-style Stage 0 cloud screening. After that, the clearest TRL 5 path is a hosted-payload pilot through Loft Orbital, with external validation focused on Y1 pilot assumptions, agronomy review, and channel plausibility.

If selected as a finalist, the immediate pre-pitch actions are a live MODIS replay demo, a first-pass Loft Orbital cost estimate, and two farmer interviews on latency and pricing assumptions.

6 Conclusion

MASFE is a credible Space-to-Soil solution: Table 2 and §3 capture latency, resolution, false-positive, SWAP, and downlink performance at the stated operating point. Released under the MIT License, it is deployable by food-security programmes without new sensing infrastructure; Mutanga et al. document 30–60% maize yield losses from grey leaf spot in Africa and note that fungicides are often sprayed uniformly because timely patch-level disease maps are unavailable [24]. Closing TRL 4 requires the hardware path laid out in §5.

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